Lecture 3: Unconstrained Optimization

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Outline

Consider a real-valued function of n variables $F \colon U \subset \mathbb{R}^n \to \mathbb{R}$. A point $\mathbf{x}^*$ is a

- **maximizer** of F on U if $F(\mathbf{x}^*) \geq F(\mathbf{x})$ for all $\mathbf{x} \in U$
- **strict maximizer** if $F(\mathbf{x}^*) > F(\mathbf{x})$ for all $\mathbf{x} \neq \mathbf{x}^* \in U$
- **local/relative maximizer** if there is a ball $B_r(\mathbf{x}^*)$ about $\mathbf{x}^*$ such that $F(\mathbf{x}^*) \geq F(\mathbf{x})$ for all $\mathbf{x} \in B_r(\mathbf{x}^*) \cap U$
- **strict local maximizer** if there is a ball $B_r(\mathbf{x}^*)$ about $\mathbf{x}^*$ such that $F(\mathbf{x}^*) > F(\mathbf{x})$ for all $\mathbf{x} \neq \mathbf{x}^* \in B_r(\mathbf{x}^*) \cap U$

Theorem 17.1

Let $F\colon U \subset \mathbb{R}^n \to \mathbb{R}^1$ be a C^1 function. If $\mathbf{x}^*$ is a local max or min of F in U and if $\mathbf{x}^*$ is an interior point of U, then

$$\frac{\partial F}{\partial x_i}(\mathbf{x}^*) = 0 \qquad \text{for } i = 1, \cdots, n.$$

Proof: for each x_i, x_i^* is an interior maximizer of $x_i \mapsto F(x_1^*, \ldots, x_{i-1}^*, x_i, x_{i+1}^*, \ldots, x_n^*)$. Use Theorem 3.3.

Outline

Definition

Definition: A quadratic form on $\mathbb{R}^k$ is a real-valued function of the form

$$Q(x_1, \cdots, x_k) = \sum_{i \le j} a_{ij} x_i x_j$$

in which each term is a monomial of degree two.

A quadratic form can be written in matrix form:

$$\mathbf{x}^T A \mathbf{x} = \begin{pmatrix} x_1 & x_2 & \cdots & x_k \end{pmatrix} \begin{pmatrix} a_{11} & \frac{1}{2}a_{12} & \cdots & \frac{1}{2}a_{1k} \\ \frac{1}{2}a_{12} & a_{22} & \cdots & \frac{1}{2}a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{2}a_{1k} & \frac{1}{2}a_{2k} & \cdots & a_{kk} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix}$$

where A is a symmetric matrix.

Definiteness

A quadratic form (or a symmetric matrix A) is

- positive definite: if $\mathbf{x}^T A\mathbf{x} > 0$ for all $\mathbf{x} \neq 0$ in $\mathbb{R}^n$. For example: $Q(x_1, x_2) = x_1^2 + x_2^2$
- negative definite: if $\mathbf{x}^T A\mathbf{x} < 0$ for all $\mathbf{x} \neq 0$ in $\mathbb{R}^n$. For example: $Q(x_1, x_2) = -x_1^2 - x_2^2$
- positive semi-definite: if $\mathbf{x}^T A\mathbf{x} \geq 0$ for all $\mathbf{x} \neq 0$ in $\mathbb{R}^n$. For example: $Q(x_1, x_2) = x_1^2 + 2x_1x_2 + x_2^2$
- negative semi-definite: if $\mathbf{x}^T A\mathbf{x} \leq 0$ for all $\mathbf{x} \neq 0$ in $\mathbb{R}^n$. For example: $Q(x_1, x_2) = -x_1^2 - 2x_1x_2 - x_2^2$
- indefinite: if $\mathbf{x}^T A\mathbf{x} > 0$ for some $\mathbf{x}$ in $\mathbb{R}^n$ and < 0 for some other $\mathbf{x}$ in $\mathbb{R}^n$. for example: $Q(x_1, x_2) = x_1^2 - x_2^2$

Definition

Let A be an $n \times n$ matrix. A $k \times k$ submatrix of A formed by deleting $n-k$ columns, say columns $i_1, i_2, \cdots, i_{n-k}$ and the same rows $i_1, i_2, \cdots, i_{n-k}$ from A is called a kth order principal submatrix of A. The determinant of a $k \times k$ principal submatrix is called a kth order principal minor of A.

Definition

Let A be an $n \times n$ matrix. The kth order principal submatrix of A obtained by deleting the last $n-k$ rows and the last $n-k$ columns from A is called kth order leading principal submatrix of A. Its determinant is called kth order *leading principal* minor of A.

For example, for a 3×3 matrix, there are three leading principal minors:

$$|a_{11}|, \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Theorem 16.1

Let A be an $n \times n$ symmetric matrix. Then,

(a) A is positive definite *if and only if* all its n leading principal minors are (strictly) positive

(b) A is negative definite *if and only if* its n leading principal minors alternate in sign as follows: $|A_1| < 0$, $|A_2| > 0$, $|A_3| < 0$, etc. That is, the kth order leading principal minor should have the same sign as $(-1)^k$.

(c) If some kth order leading principal minor of A is nonzero but does not fit either of the above two sign patterns, then A is indefinite. This case occurs when A has a negative kth order leading principal minor for an *even* integer k; or when A has a negative kth order leading principal minor and a positive ℓth order leading principal minor for two distinct *odd* integers k and ℓ.

This test may fail for a symmetric matrix when some leading principal minor is 0 while the nonzero ones fit the pattern in either (a) or (b).

Theorem 16.2

Let A be an $n \times n$ symmetric matrix. Then

- A is positive semidefinite *if and only if* **every** principal minor of A is ≥ 0
- A is negative semidefinite *if and only if* **every** principal minor of odd order is ≤ 0 and every principal minor of even order is ≥ 0.

Definiteness of Diagonal Matrices

- The simplest $n \times n$ symmetric matrix is the diagonal matrix, which corresponds to the simplest quadratic form $a_1 x_1^2 + \cdots a_n x_n^2$
- It is positive definite $\iff$ all the a_i's are positive and negative definite $\iff$ all the a_i's are negative
- It is positive semidefinite $\iff$ all $a_i \geq 0$ and negative semidefinite $\iff$ all $a_i \leq 0$
- If there are two a_i's of opposite signs, it will be indefinite
- If some $a_j = 0$, it is not definite

Definiteness and Optimality

- Fact: determining the definiteness of a quadratic form Q is equivalent to determining whether $\mathbf{x} = \mathbf{0}$ is a max, a min, or neither for the real-valued function Q.
- Figures 16.2–16.6

Outline

Second Order Conditions

- A point x^* is a **critical point** if $DF(\mathbf{x}^*) = 0$
- To determine if a critical point is a max or min, we need to use a condition on the second derivatives of F
- **Hessian** of F:

$$\begin{pmatrix} \frac{\partial^2 F}{\partial x_1^2} & \cdots & \frac{\partial^2 F}{\partial x_n \partial x_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 F}{\partial x_1 \partial x_n} & \cdots & \frac{\partial^2 F}{\partial x_n^2} \end{pmatrix}$$

Theorem 17.2 (Sufficient Conditions)

Let $F\colon U \subset \mathbb{R}^n \to \mathbb{R}^1$ be a C^2 function whose domain is an open set U in $\mathbb{R}^n$. Suppose $\mathbf{x}^*$ is a critical point of F:

(a) if the Hessian $D^2F(\mathbf{x}^*)$ is a negative definite symmetric matrix, then $\mathbf{x}^*$ is a strict local max of F;

(b) if the Hessian $D^2F(\mathbf{x}^*)$ is a positive definite symmetric matrix, then $\mathbf{x}^*$ is a strict local min of F;

(c) if the Hessian $D^2F(\mathbf{x}^*)$ is indefinite, then $\mathbf{x}^*$ is neither a local max nor a local min of F.

- Equivalent statements based on analytical characterization of positive definite and negative definite matrices: Theorem 17.3-17.5
- A critical point of F for which the Hessian $D^2F(\mathbf{x}^*)$ is indefinite is called a **saddle point**
- A saddle point is a min of F in some directions and a max in other directions
- Example: $F(x_1, x_2) = x_1^2 - x_2^2$

Theorem 17.6

Let $F\colon U \subset \mathbb{R}^n \to \mathbb{R}^1$ be a C^2 function. Suppose $\mathbf{x}^*$ is an interior point of U and a local max/min of F. Then

$$DF(\mathbf{x}^*) = \mathbf{0}$$

and $D^2F(\mathbf{x}^*)$ is negative/positive **semi**definite.

Theorem 17.7

Let $F\colon U \subset \mathbb{R}^n \to \mathbb{R}^1$ be a C^2 function of n variables. Suppose $\mathbf{x}^*$ is an interior point of U and a local max/min of F.

(a) if $\mathbf{x}^*$ is a local min of F, then $\frac{\partial F}{\partial x_i}(\mathbf{x}^*) = 0$ for $i = 1, \cdots, n$ and all the principal minors of the Hessian $D^2F(\mathbf{x}^*)$ are ≥ 0.

(b) if $\mathbf{x}^*$ is a local max of F, then $\frac{\partial F}{\partial x_i}(\mathbf{x}^*) = 0$ for $i = 1, \cdots, n$ and all the principal minors of the Hessian $D^2F(\mathbf{x}^*)$ of odd order are ≤ 0 and all the principal minors of the Hessian $D^2F(\mathbf{x}^*)$ of even order are ≥ 0.

Theorem 17.8

Let $F\colon U \to \mathbb{R}^1$ be a C^2 function whose domain is a convex open subset U of $\mathbb{R}^n$.

(a) The following three conditions are equivalent:
- (i) F is a concave function on U; and
- (ii) $F(\mathrm{y}) - F(\mathrm{x}) \leq DF(\mathrm{x})(\mathrm{y} - \mathrm{x})$ for all $\mathrm{x}, \mathrm{y} \in U$; and
- (iii) $D^2F(\mathrm{x})$ is negative semidefinite for all $\mathrm{x} \in U$

(b) The following three conditions are equivalent:
- (i) F is a convex function on U; and
- (ii) $F(\mathrm{y}) - F(\mathrm{x}) \geq DF(\mathrm{x})(\mathrm{y} - \mathrm{x})$ for all $\mathrm{x}, \mathrm{y} \in U$; and
- (iii) $D^2F(\mathrm{x})$ is positive semidefinite for all $\mathrm{x} \in U$

(c) If F is concave on U and $DF(\mathbf{x}^*) = \mathbf{0}$ for some $\mathbf{x}^* \in U$, then $\mathbf{x}^*$ is a global max of F on U.

(d) If F is convex on U and $DF(\mathbf{x}^*) = \mathbf{0}$ for some $\mathbf{x}^* \in U$, then $\mathbf{x}^*$ is a global min of F on U.

- A monopolist faces two distinct and separated markets (e.g., a domestic market and a foreign market), each with its own demand function
  - supply: Q_i
  - inverse demand function: $P_i = G_i(Q_i)$
  - revenue: $Q_i G_i(Q_i)$
  - production costs: $C(Q_1 + Q_2)$
- Profit: $F(Q_1, Q_2) = Q_1 G_1(Q_1) + Q_2 G_2(Q_2) - C(Q_1 + Q_2)$
- Suppose that the firm will produce a positive amount for each market
- Problem: compute the maxima of the profit function F in the interior of the positive quadrant

- Then we have
$$\frac{d(Q_1 G_1(Q_1))}{dQ_1} = \frac{d(Q_2 G_2(Q_2))}{dQ_2} = C'(Q_1 + Q_2)$$
- The marginal revenue in *each* market = marginal cost of the total output

- $G(Q_1) = 50 - 5Q_1$
- $G(Q_2) = 100 - 10Q_2$
- $C(Q) = 90 + 20Q$
- In order to maximize profits, how much should the monopolist produce for each market?

- This discriminating monopolist's profit function is

$$F(Q_1, Q_2) = Q_1(50 - 5Q_1) + Q_2(100 - 10Q_2) - (90 + 20(Q_1 + Q_2))$$

- The critical point of F satisfy:

$$\frac{\partial F}{\partial Q_1} = 50 - 10Q_1 - 20 = 0$$
$$\frac{\partial F}{\partial Q_2} = 100 - 20Q_2 - 20 = 0$$

Hence, $Q_1 = 3$ and $Q_2 = 4$

- Now check the second order conditions: $F_{Q_1Q_1} = -10$, $F_{Q_2Q_2} = -20$, $F_{Q_1Q_2} = F_{Q_2Q_1} = 0$
- The first order leading principal minor of $D^2F(3,4)$ is -10 and second order leading principal minor is 200.
- Therefore, F is a concave function and the point $(3,4)$ is a maximizer